

**LESSON 117 (1.0 GROUND and 1.3 DUAL)****Lesson Objective**

During this lesson, the student will gain proficiency with emergency procedures and normal traffic pattern procedures. The student will be introduced to the no flap and power off approach as well as the rejected takeoff.

**Briefing**

- Review homework assigned in the previous lesson.
  - Quiz the student on emergency procedures and checklists.
  - Quiz the student on towered operations.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

**References**

- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
  - Chapter 05 - Aerodynamics of Flight.
  - Chapter 12 - Weather Theory.
- Aeronautical Information Manual. ([AIM](#))
  - Chapter 06 - Emergency Procedures.
  - Chapter 07 - Safety of Flight.
    - Section 04 - Wake Turbulence.
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
  - Chapter 08 - Airport Traffic Patterns.
  - Chapter 09 - Approaches and Landings.

**Lesson Content (Ground)****Runway Incursion Avoidance**

- Define a runway incursion.
- Highlight the 4 categories:
  - Category A: serious incident in which a collision was narrowly avoided.
  - Category B: incident in which separation decreases and there is a significant potential for collision.
  - Category C: incident characterized by ample time and/or distance to avoid a collision.
  - Category D: incident that meets the definition of runway incursion such as incorrect presence of a single vehicle/person/aircraft on the protected area of a surface designated for the landing and takeoff of aircraft but with no immediate safety consequences.
- Explain how to avoid a runway incursion:
  - Always brief taxi instructions.
  - Be familiar with the airport layout.

## Enhanced Private Pilot Certification Course

- Know where hold-short lines are located.
- Do not cross a hold-short line without obtaining clearance.
- Ensure the runway is clear of vehicle/person/aircraft before taking-off and landing.

### Windshear

- Define what wind shear is for the student.
  - Windshear is a significant change in wind speed and/or direction.
- Low Level Wind Shear (LLWS) occurs below 2000' AGL and poses a significant threat to all aircraft.
- Draw three situations to help the student visualize the impacts of wind shear.
  - First, draw an aircraft on final in a no wind condition.
    - Label indicated airspeed and ground speed.
  - Second, draw that aircraft encountering a 10kt head wind.
    - Label the indicated airspeed increasing and the ground speed decreasing.
    - Also point out how a pilot would react to this:
      - Increase pitch to reduce airspeed to approach speed and continue approach at 63kts.
  - Third, draw that aircraft with a 10kt headwind shearing to a tailwind.
    - Label the 20kt decrease in indicated airspeed and 20kt increase in ground speed.
    - What does this leave our indicated airspeed at?
    - What is  $V_{s0}$ ?

### Wake Turbulence

- Explain how wake turbulence is created:
  - Pressure differential → low pressure above and high pressure below the wing.
  - Air moves from the high-pressure area under the wing to the low-pressure area over the wing.
    - It goes around the wing tip and then trails behind.
    - As air curls around the tip, it combines with the wash to form a fast spinning trailing vortex.
    - Right wing tip - counterclockwise.
    - Left wing tip - clockwise.
- Explain how to avoid wake turbulence:
  - Vortices are greatest when an aircraft is in a heavy, slow, and clean configuration.
    - Avoid flying through another aircraft's flightpath.
    - Rotate prior to proceeding aircraft.
    - Avoid following aircraft on similar flight paths.
    - Land past the touchdown point of the other aircraft.
  - Wind is an important factor to wake turbulence.

### Radio Communications Failure Procedures

- The first step is always to troubleshoot:
  - Double check the volume.
    - Audio panel.
    - Comm volume.
    - Headset volume.
  - Check to make sure the headset is plugged in.
  - Ensure the frequency is set correctly.
  - Attempt using the alternate communications radio (com1/com2).
  - Attempt to call the tower with a cell phone.
- If troubleshooting is unsuccessful, squawk 7600.
- If close to a non-towered airport, proceed there.
- If not, circle over the towered airport at 500' above traffic pattern altitude.
- Review light gun signals.

### System/Equipment Malfunction

- Give the student different malfunction scenarios.
  - Loss of oil pressure, high oil temperature, carburetor icing, hole in the exhaust shroud (while using cabin heat), etc.

### No Flap Approach

- Ask the student why we use flaps on approach.
  - Lower stall speed (increase in camber, increase in lift coefficient) ( $V_{S0}$  vs  $V_{S1}$ ).
  - Increase in lift.
  - Increase in induced drag.
  - Steeper angle of descent without increasing airspeed.
- Flying a no flap approach:
  - Higher nose attitude.
  - Losing altitude will be more difficult.
  - Wider traffic pattern.
  - Sight picture will be different (highlight this during the flight).
  - The airplane will tend to float more during the round out and flare phases due to the lack of induced drag.
  - A longer ground roll will be experienced.
- When do we do a no flap approach:
  - Flap malfunction.
  - Very windy/gusty conditions.
  - Flying into a busy airport (max forward speed).

### Power Off Approach

- In the event of an engine failure in the traffic pattern or in close proximity of an airport, a power off approach will be required.

## Enhanced Private Pilot Certification Course

- Contact ATC when able.
  - Pitch for  $V_{\text{glide}}$ .
  - Highlight how the winds will drastically affect when to turn base and final.
  - Extending the flaps to 10 degrees helps maintaining a stable traffic pattern, even with an engine failure.
  - Flaps 25 should not be extended until the runway can be made for sure.
  - Flaps 40 should only be used if coming in too high as this drastically increases drag.
  - Once on the runway, turn on the nearest taxiway.

### Engine Failure after Takeoff

- An engine failure after takeoff requires an immediate response.
- Review the takeoff briefing:
  - If sufficient runway is remaining, land on the remaining runway.
  - If no sufficient runway is remaining, make shallow turns, maintain a safe airspeed, and find the best place to land (Ex: if taking off from runway 5 and an engine failure happens, try to turn left to land on 27L).
  - If there is sufficient altitude (above 1,000ft), turn back around to the airport and attempt an engine restart.
    - Review the engine power loss in flight checklist.
- Contact ATC.

### Rejected Takeoff

- A rejected takeoff can be due to many factors, such as:
  - Engine failure.
  - High oil pressure/temperature.
  - Low RPM.
  - A sick passenger.
  - Flight control failure.
  - Instrument failure.
  - System failure.
- In the event of a rejected takeoff:
  - Reduce the power to idle.
  - Apply breaks and aerodynamical breaking.
  - Exit the runway if able.
  - Contact ATC.

### Go/No-Go Decision

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.

**Enhanced Private Pilot Certification Course**

- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

**Lesson Content (Dual)****Preflight Inspection**

- Ensure that the student conducted a thorough preflight inspection.

**Visual Scanning and Collision Avoidance**

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

**Normal/Crosswind Takeoff**

- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.

**Normal Traffic Pattern and Powered Approach**

- The student will be in charge of all of the radio calls with tower and ground.
  - Emphasize the use of saying "say again" if the instructions are not clear.
  - Emphasize the use of saying "unable" if the instructions are not safe or cannot be followed.
- Traffic patterns are to be completed by the student with minimal instructor guidance.

**No Flap Approach and Landing**

- On one of the traffic patterns, have the student conduct a no flap approach and landing.
- The speeds on the base and final legs should be the same.
- Ensure that the student understands that a wider traffic pattern may be required.
- The student should understand that excessive floating may happen while rolling out and flaring.

**Power Off Approach and Landing**

- While on the downwind, bring the power to idle.
- Ensure that the student pitches for 73 knots.
- Guide the student through when to turn base/final depending on the wind (if needed).
- The student must be able to land on the first third of the runway.

**Engine Failure After Takeoff**

- Simulate an engine failure on takeoff.

**Enhanced Private Pilot Certification Course**

- Bring the power to idle momentarily.
- At this point, the student should talk through what they would do while the instructor applies power again and continues with the climb out.

**Forward Slips to a Landing**

- On one of the traffic patterns, have the student purposely come in high.
- Guide the student through conducting a forward slip.
  - Power should be at idle.
  - Pitch for 63 knots.
  - Ailerons into the wind.
  - Rudder opposite.

**Rejected Takeoff**

- Ask ATC for a short delay on the runway.
- As the student adds full power and gains airspeed, bring the throttle to idle to simulate an engine failure.
  - Ensure the student performs the correct procedures.

**Normal/Crosswind Landing**

- The student should conduct the landing.
- By this lesson, the student should be able to conduct a safe landing with no instructor guidance.

**Go-Around/Rejected Landing**

- Have the student conduct a go-around during one of the traffic patterns.
- Ensure that the correct procedure is conducted.

**Postflight Procedures**

- Ensure that the student conducts a thorough postflight inspection.

**Post-Brief**

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 118.
- Assign homework:
  - Give the student a TOLD card problem. This TOLD card will be used during the 118 ground.
  - Review for the mock!

**Completion Standards**

This lesson is complete when the student is able to:

1. Demonstrate proper sequential checklist procedures for all normal and emergency procedures without instructor guidance.
2. Conduct no flap, glide approach and slips to a landing with instructor guidance.
3. Conduct a rejected takeoff with instructor guidance.
4. Maintain the recommended traffic pattern altitude  $\pm 150$  ft, recommended airspeed  $+5/-0$  knots and correct for wind drift without instructor guidance.
5. Perform normal/crosswind takeoff and landings without instructor guidance.